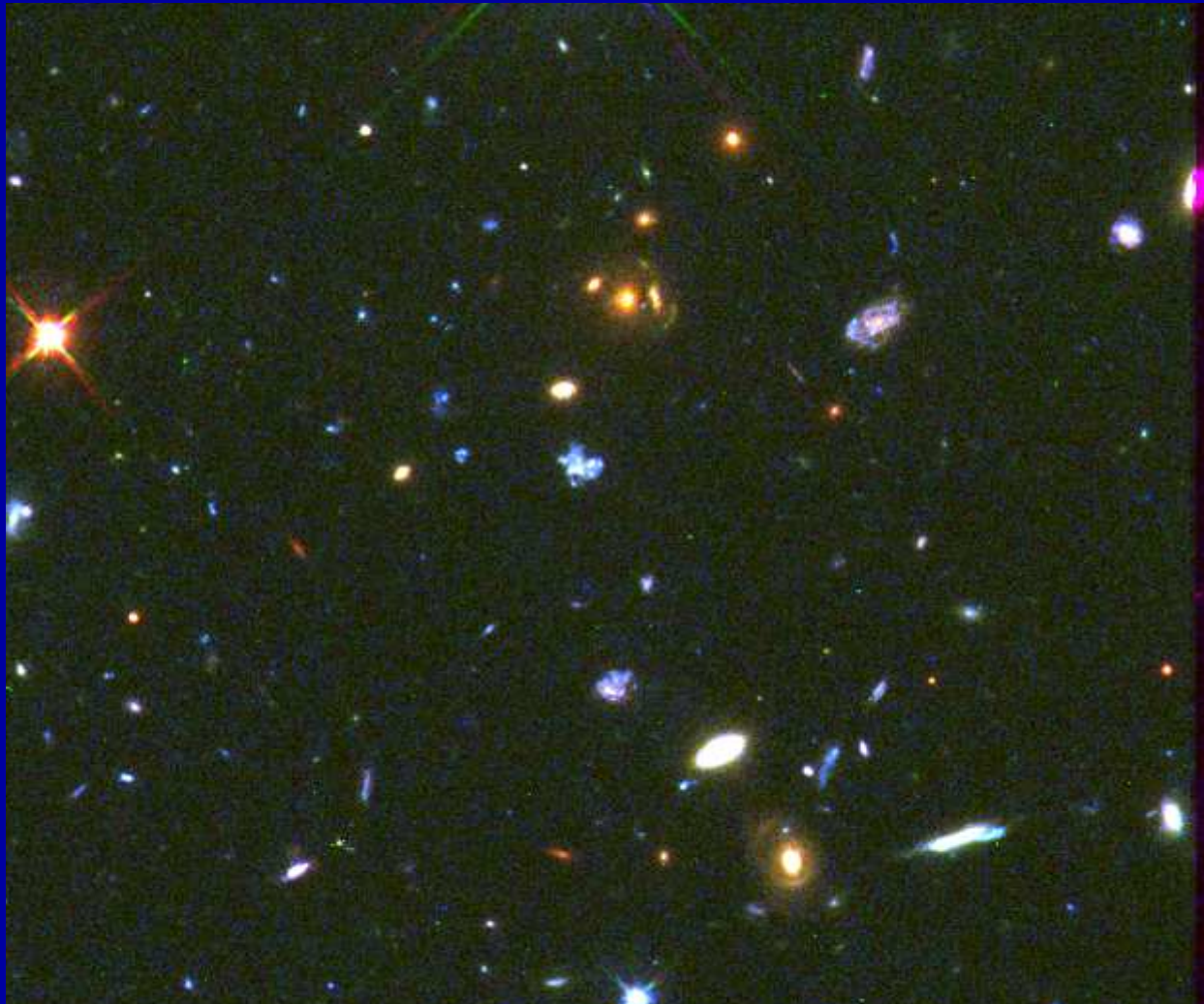


Using Hubble Space Telescope Galaxy Classification Software to Find Diabetic Type 2 in an Early Stage

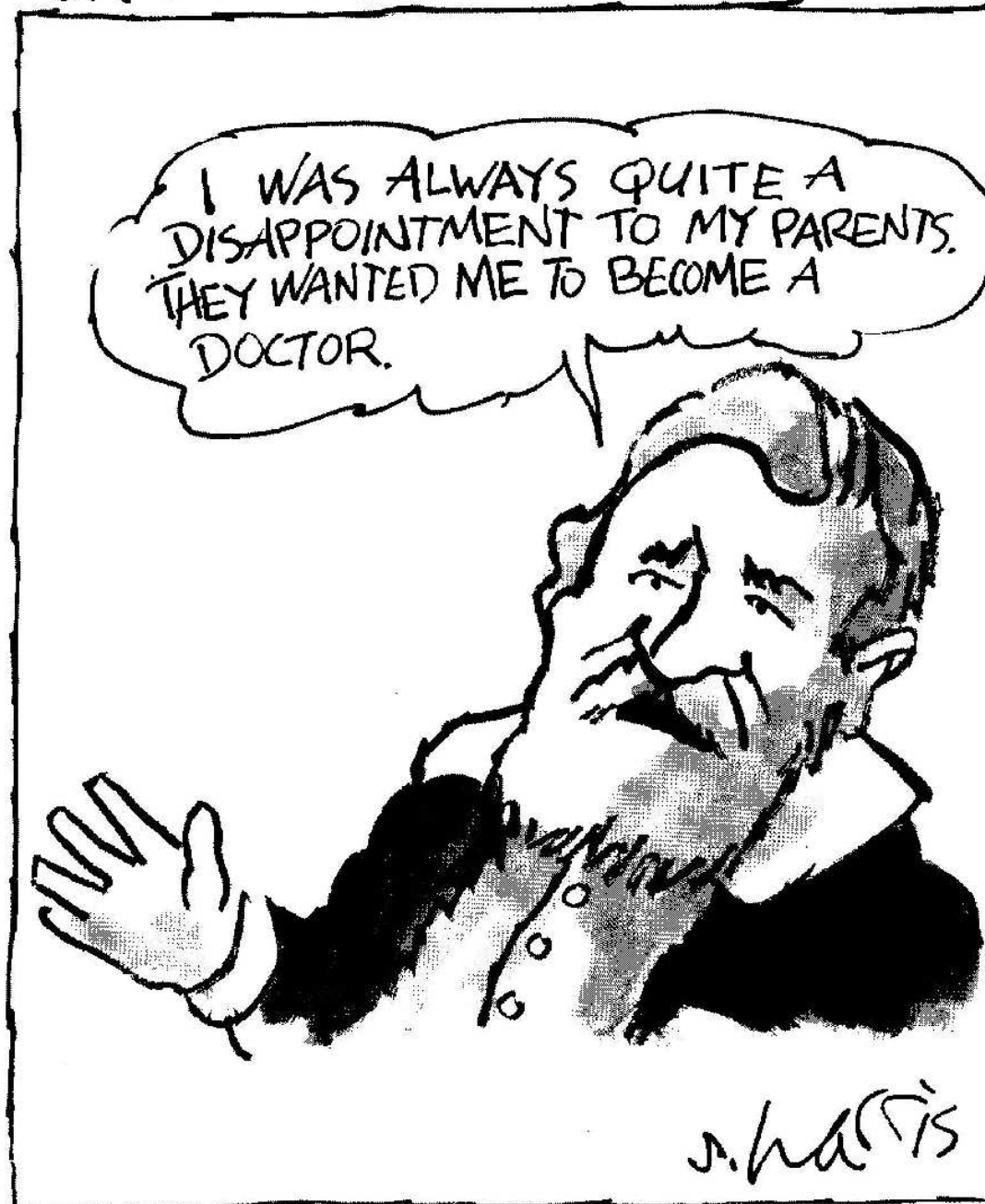
Rogier Windhorst (Physics & Astronomy, ASU)

L. Burnett, J. Brower, A. Casano, D. Cox, J. Olson (ASU), R. & J. Herman (Banner Health), K. Tamura, V. Taylor (ASU)



Palm Grant proposal review — Banner Health — Phoenix, Aug. 30, 2007

GALILEO'S BURDEN



Why I personally got involved in medical research ... (true story!)

Outline

(1) Summary of the NASA Hubble Space Telescope connection

- (1a) NASA missions to study Cosmic Dawn, First Light, Dark Ages
- (1b) NASA Hubble Space Telescope image analysis

(2) Using Galaxy Classification Software to Find Diabetic Type 2 in an Early Stage

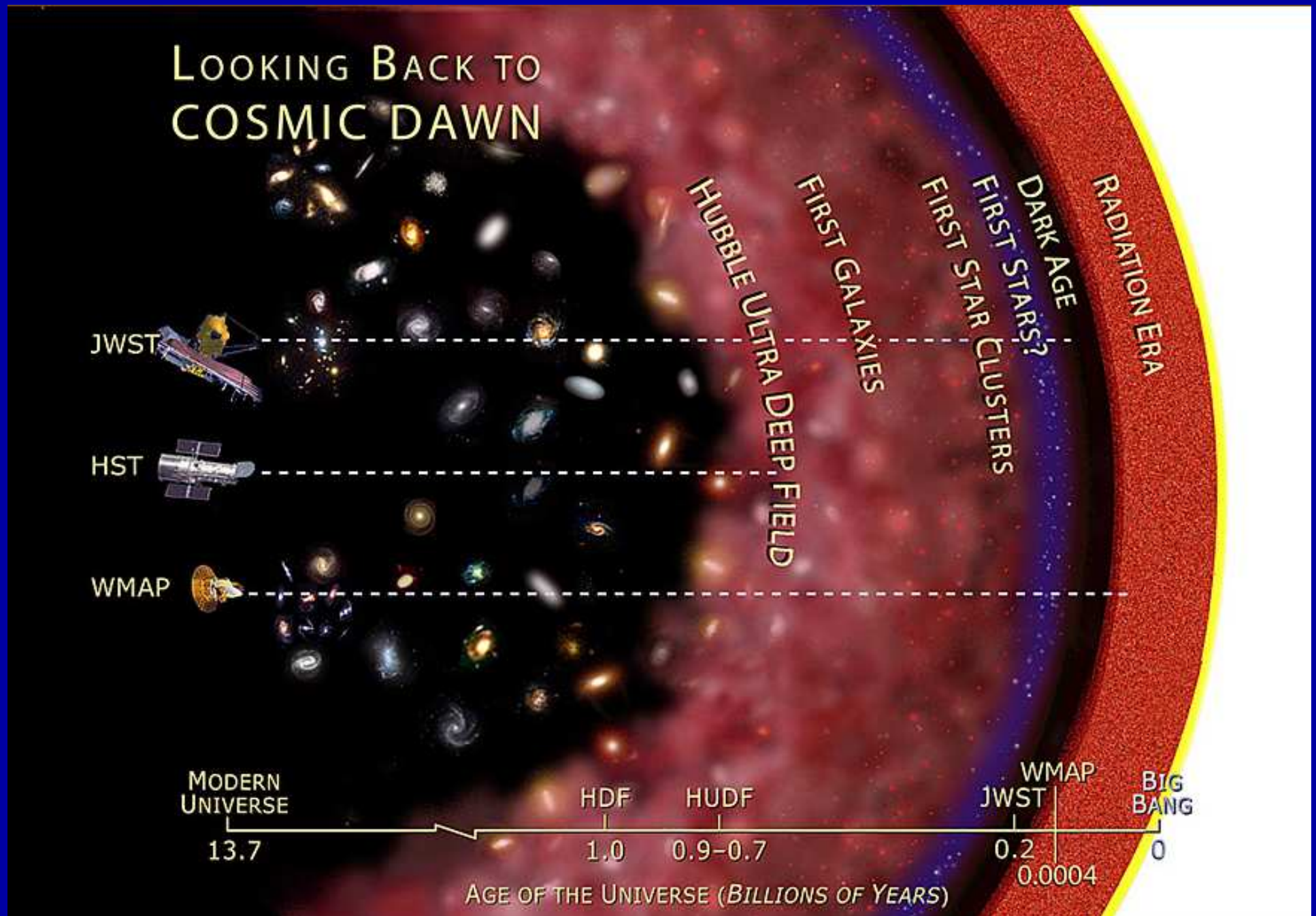
- (2a) The Galaxy Detection, Deblending, and Classification method
- (2b) Application to C-fiber Skin-biopsies to Normal and Obese Patients

(3) Results

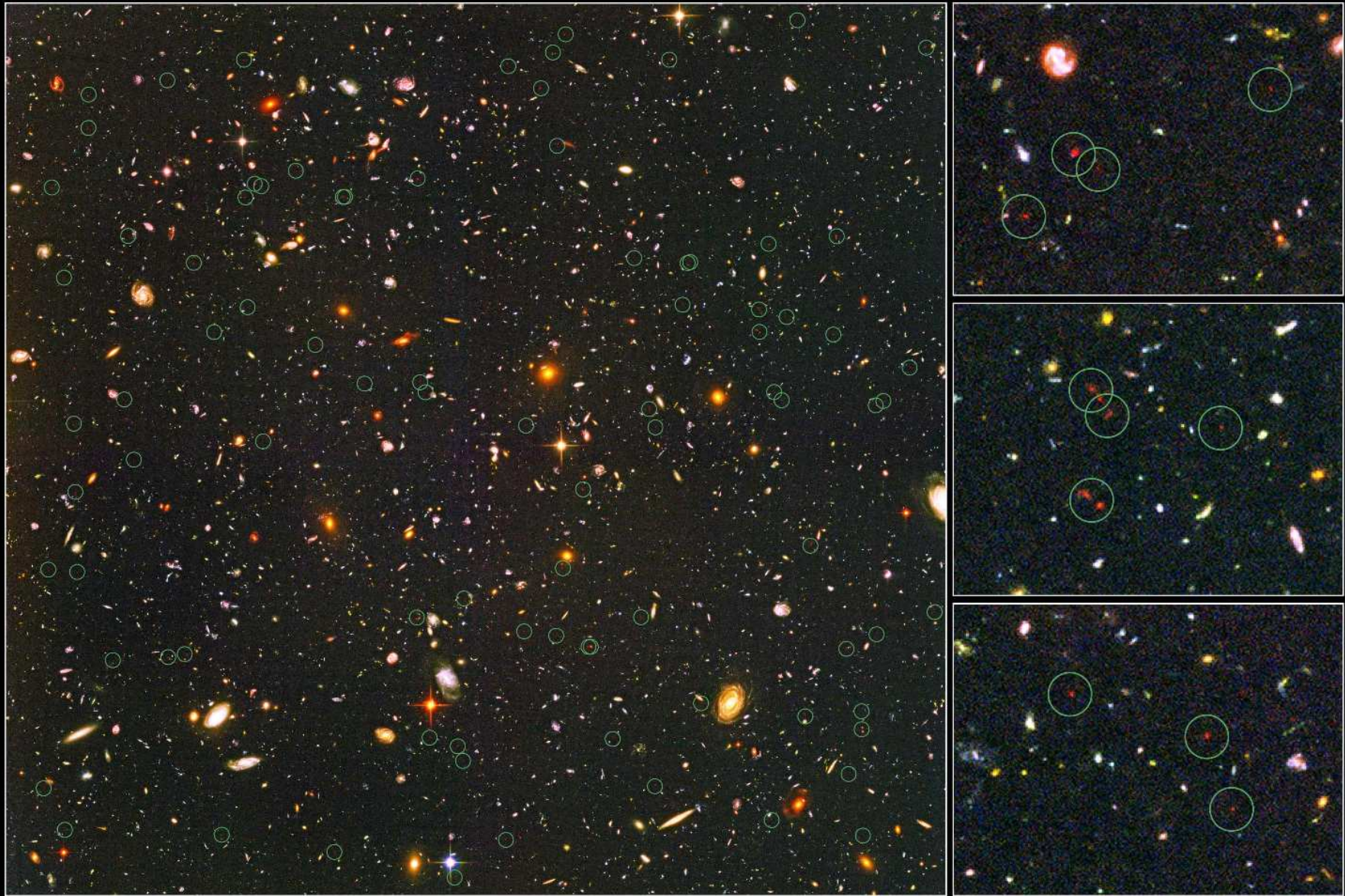
(4) Conclusions

Sponsored by NASA/ASU/Banner Health

(1a) NASA missions that study Cosmic Dawn, First Light, & the Dark Ages



Looking back in time $\Rightarrow$ deal with many overlapping galaxy images!



Distant Galaxies in the Hubble Ultra Deep Field
Hubble Space Telescope • Advanced Camera for Surveys

NASA, ESA, R. Windhorst (Arizona State University) and H. Yan (Spitzer Science Center, Caltech)

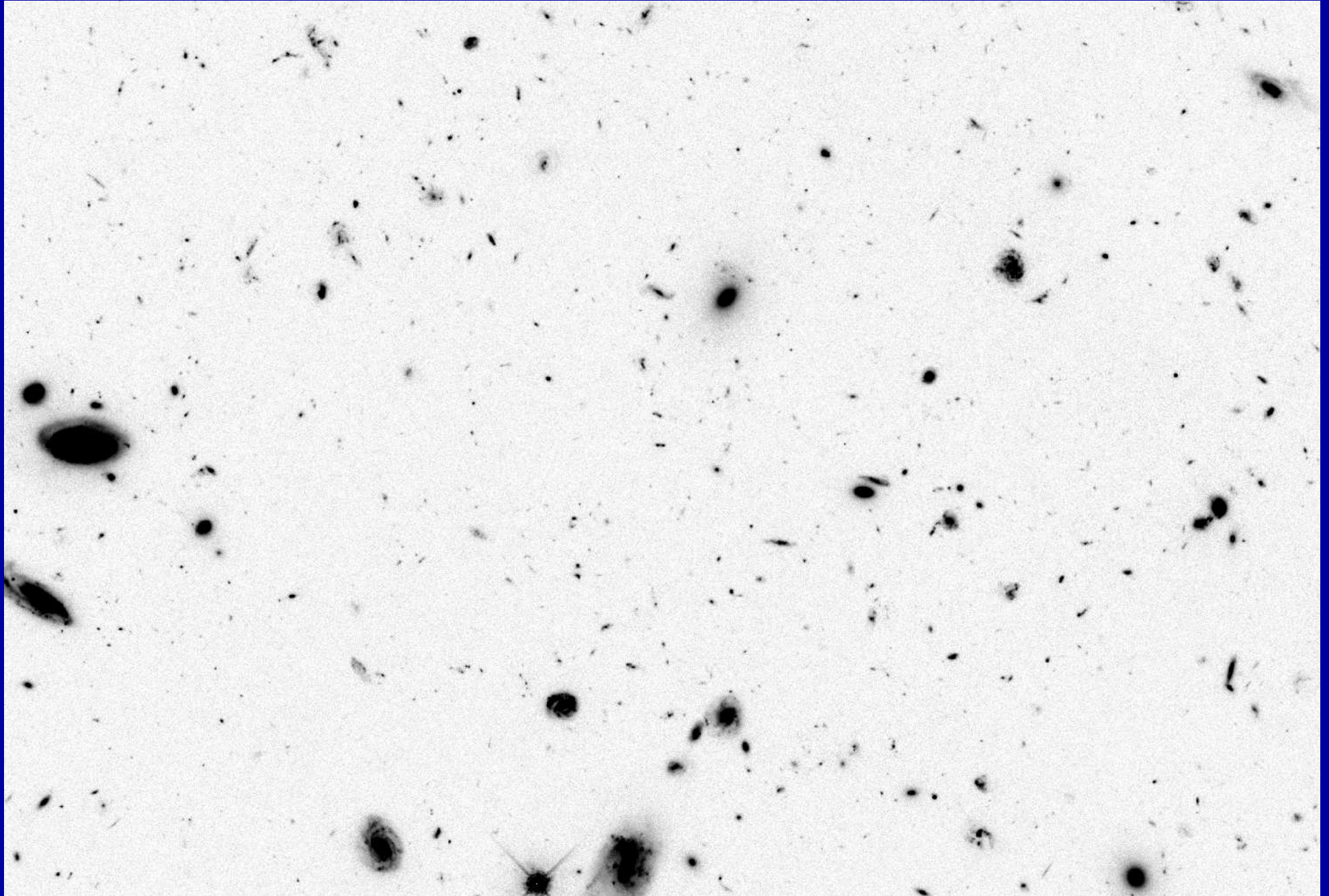
STScI-PRC04-28

A Recent Study of Cosmic Dawn with the Hubble Space Telescope



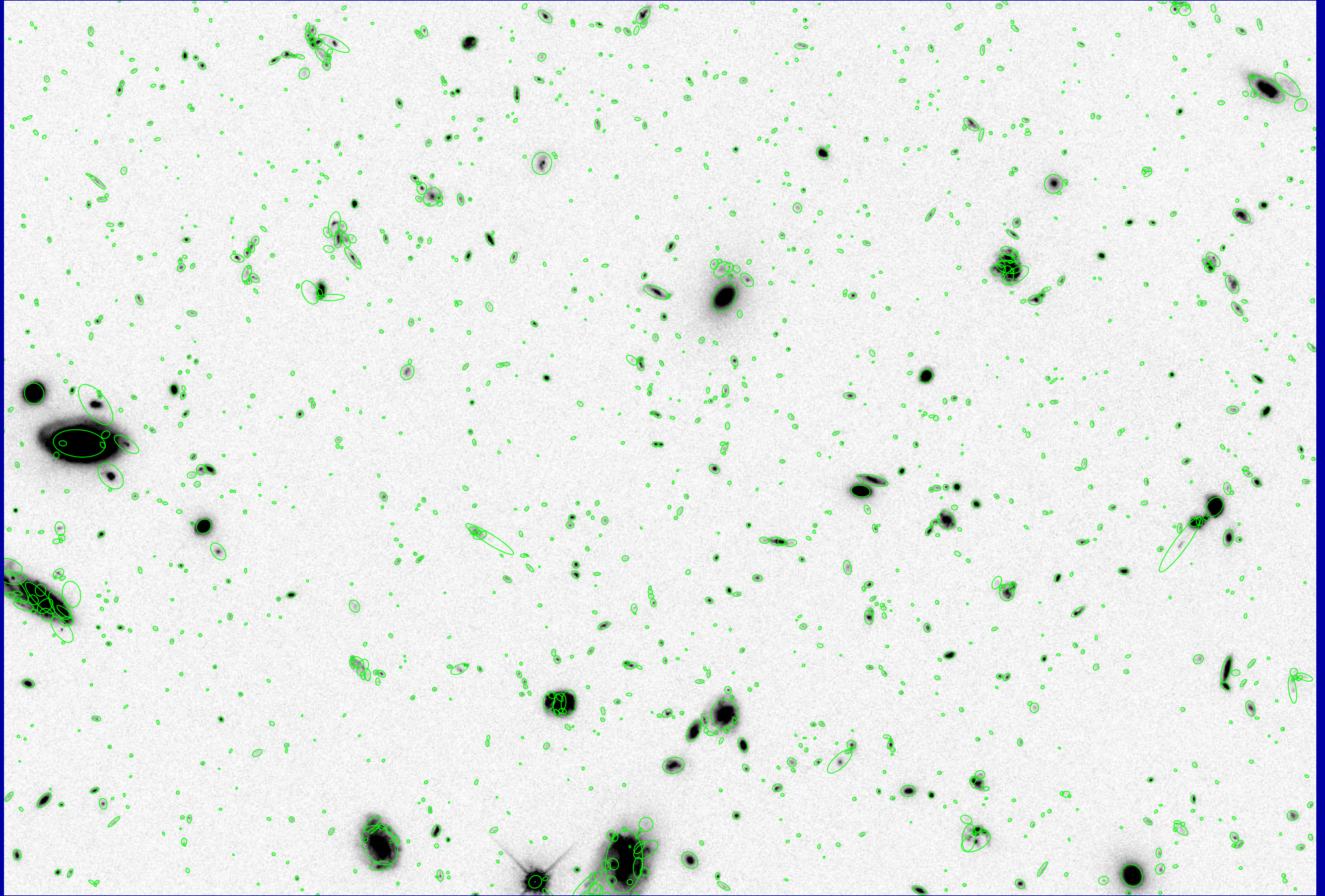
240 hour image of the Hubble UltraDeep Field: 10^4 galaxies $\lesssim 10^{10} \times$
fainter than unaided eye can see, in area smaller than looking through straw!

(1b) NASA Hubble Space Telescope image analysis



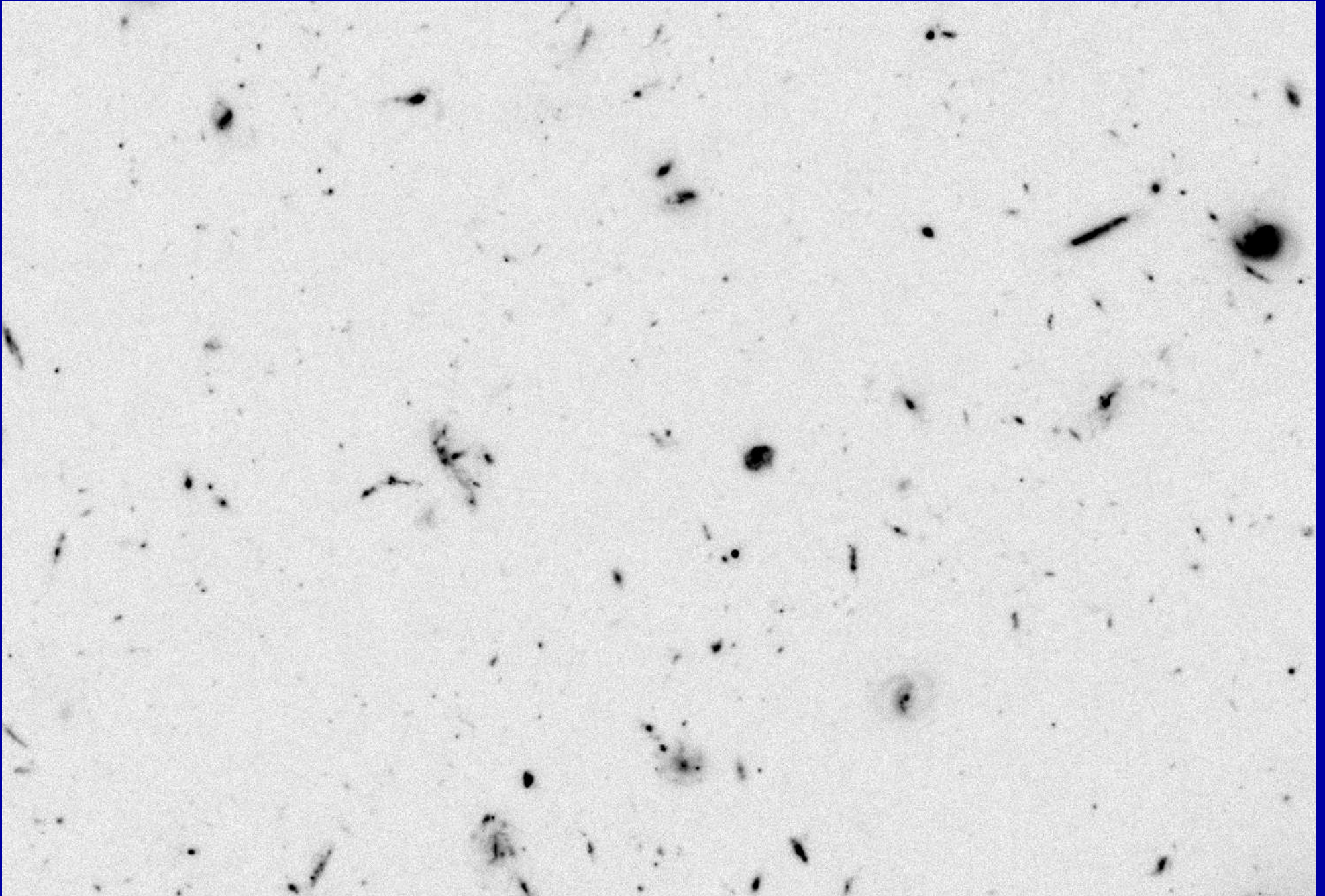
The deepest Hubble images start showing significant galaxy overlap.

(1b) NASA Hubble Space Telescope image analysis



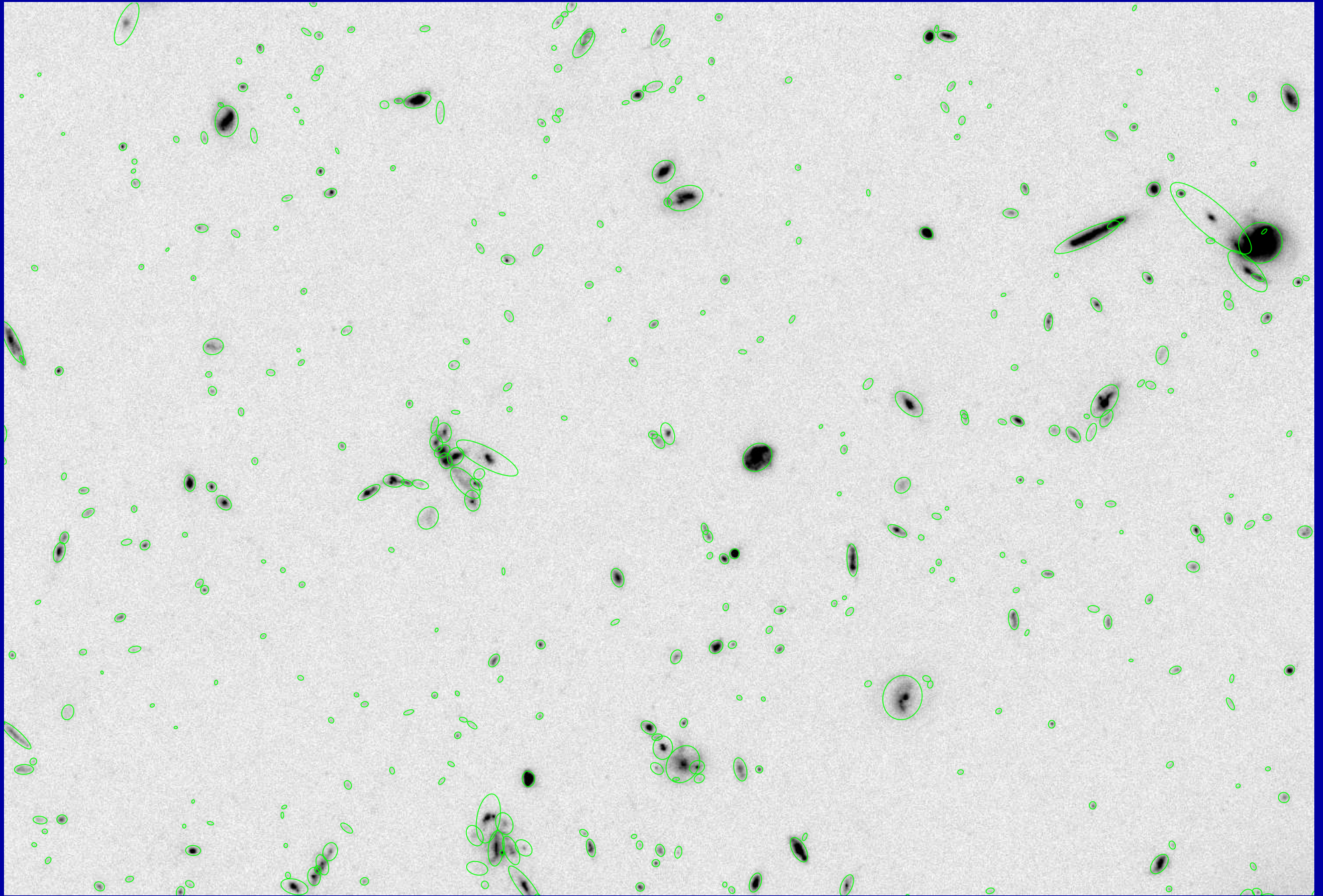
to address crowding, the source-extractor code iteratively deblends the images.

(2a) The Galaxy Detection, Deblending, and Classification method



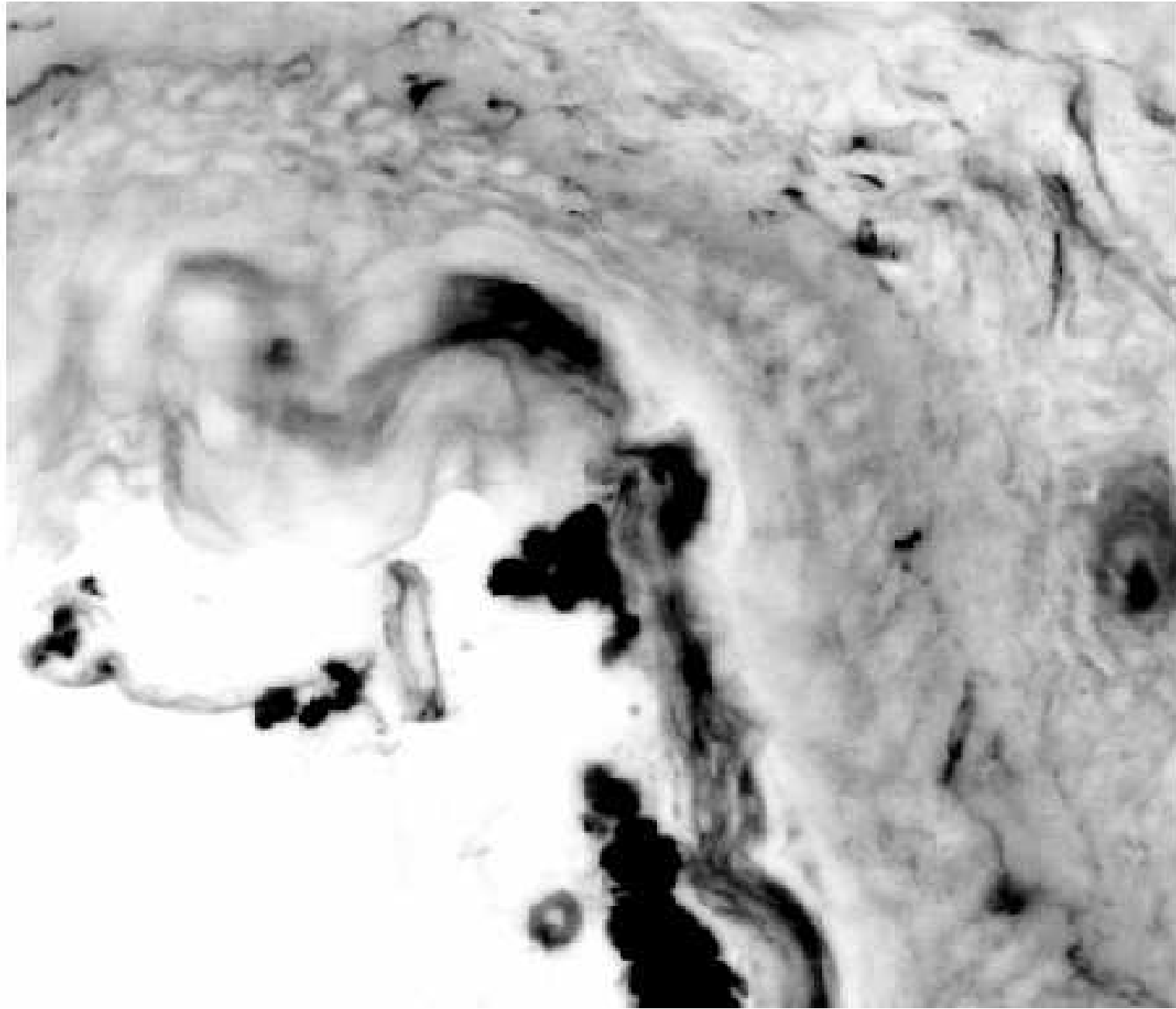
Very complex filamentary galaxies are seen in early stages of formation.

(2a) The Galaxy Detection, Deblending, and Classification method



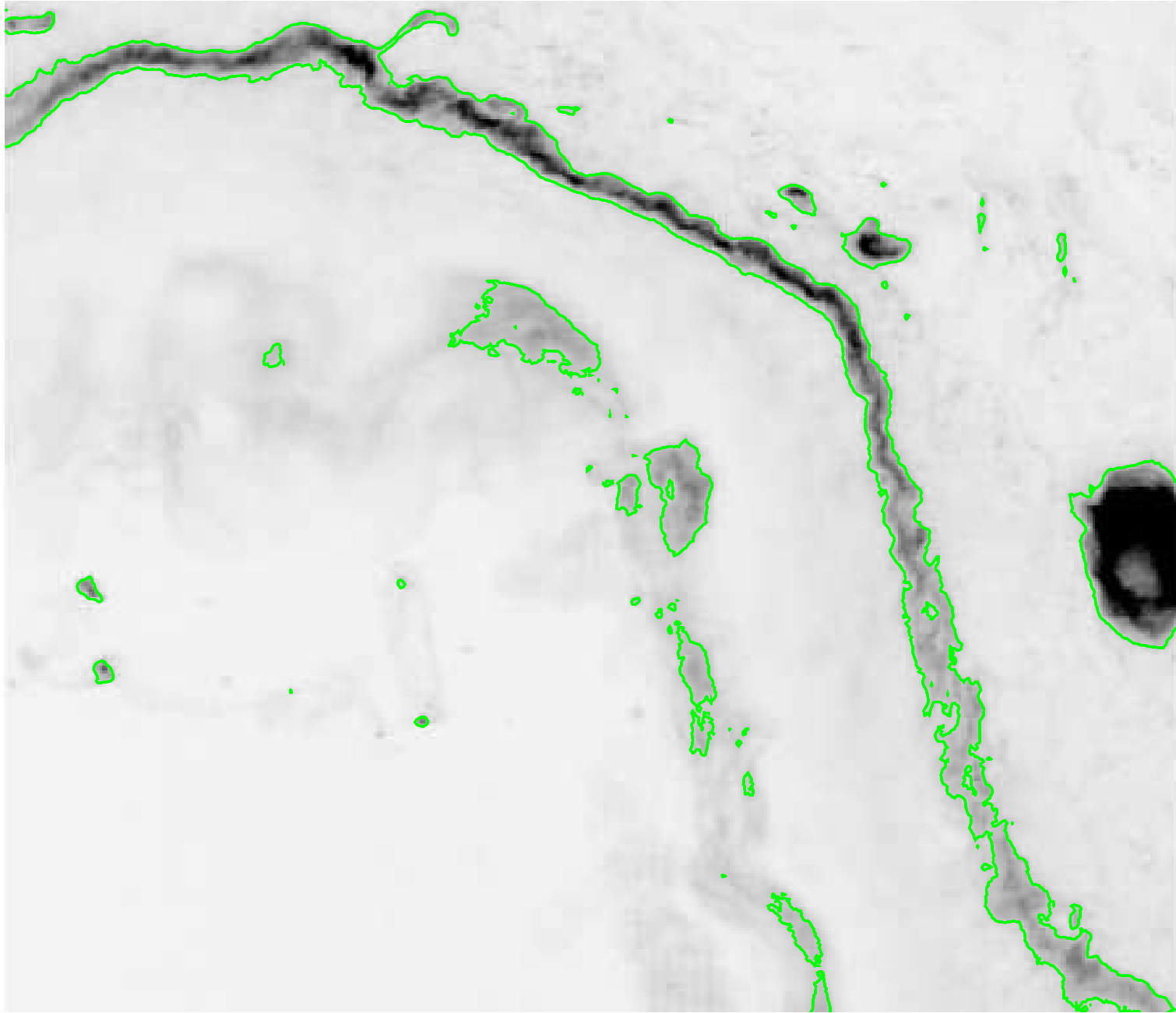
The Source-Extractor software deblends these in chains of elliptical isophotes.

(2b) Application to C-fiber Skin-biopsies in Normal and Obese Patients



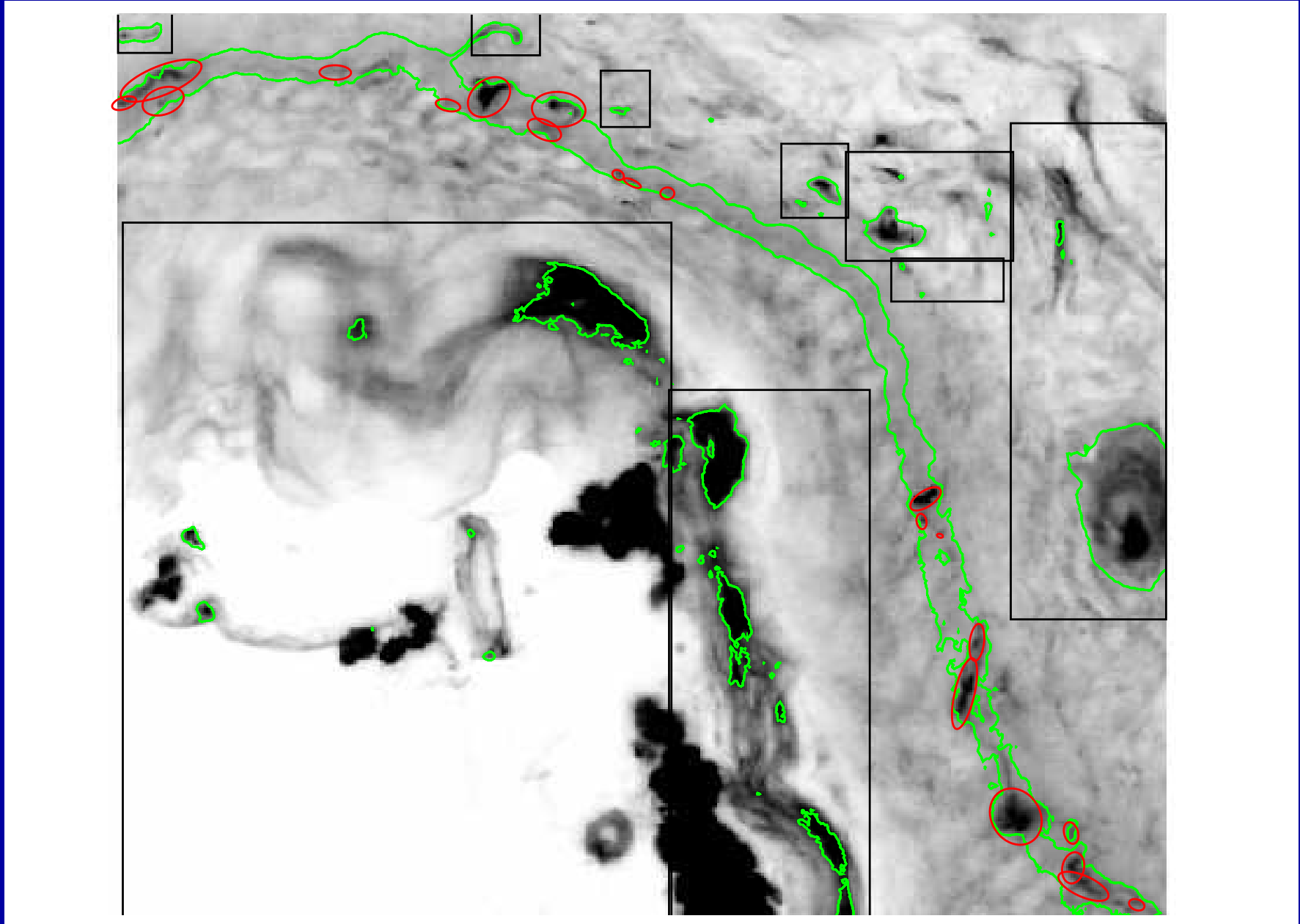
A Stained Confocal Microscope image of a Skin-Biopsy is digitized.

(2b) Application to C-fiber Skin-biopsies in Normal and Obese Patients



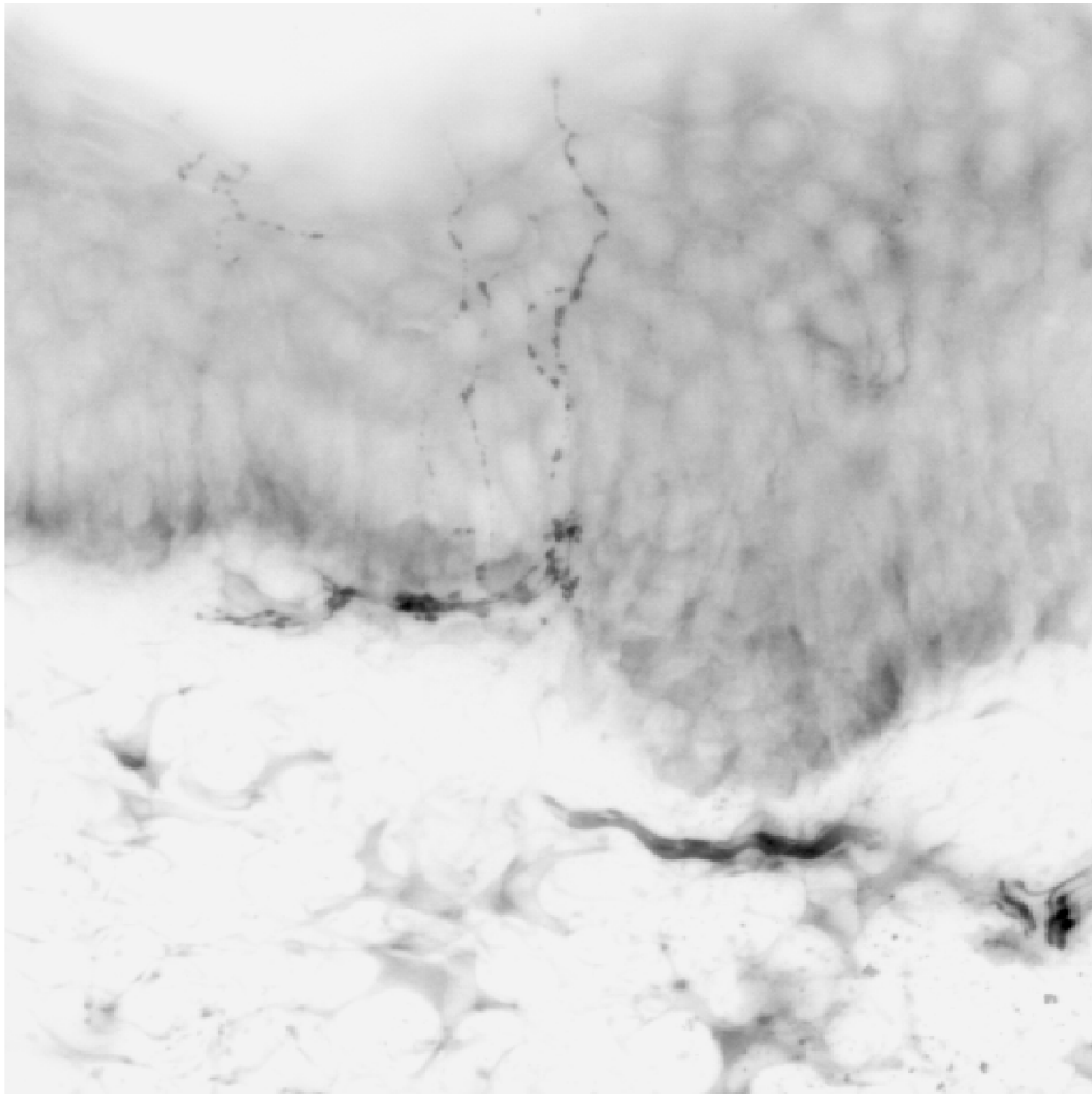
Different staining allows us to define the membrane and other parts of skin.

(2b) Application to C-fiber Skin-biopsies in Normal and Obese Patients



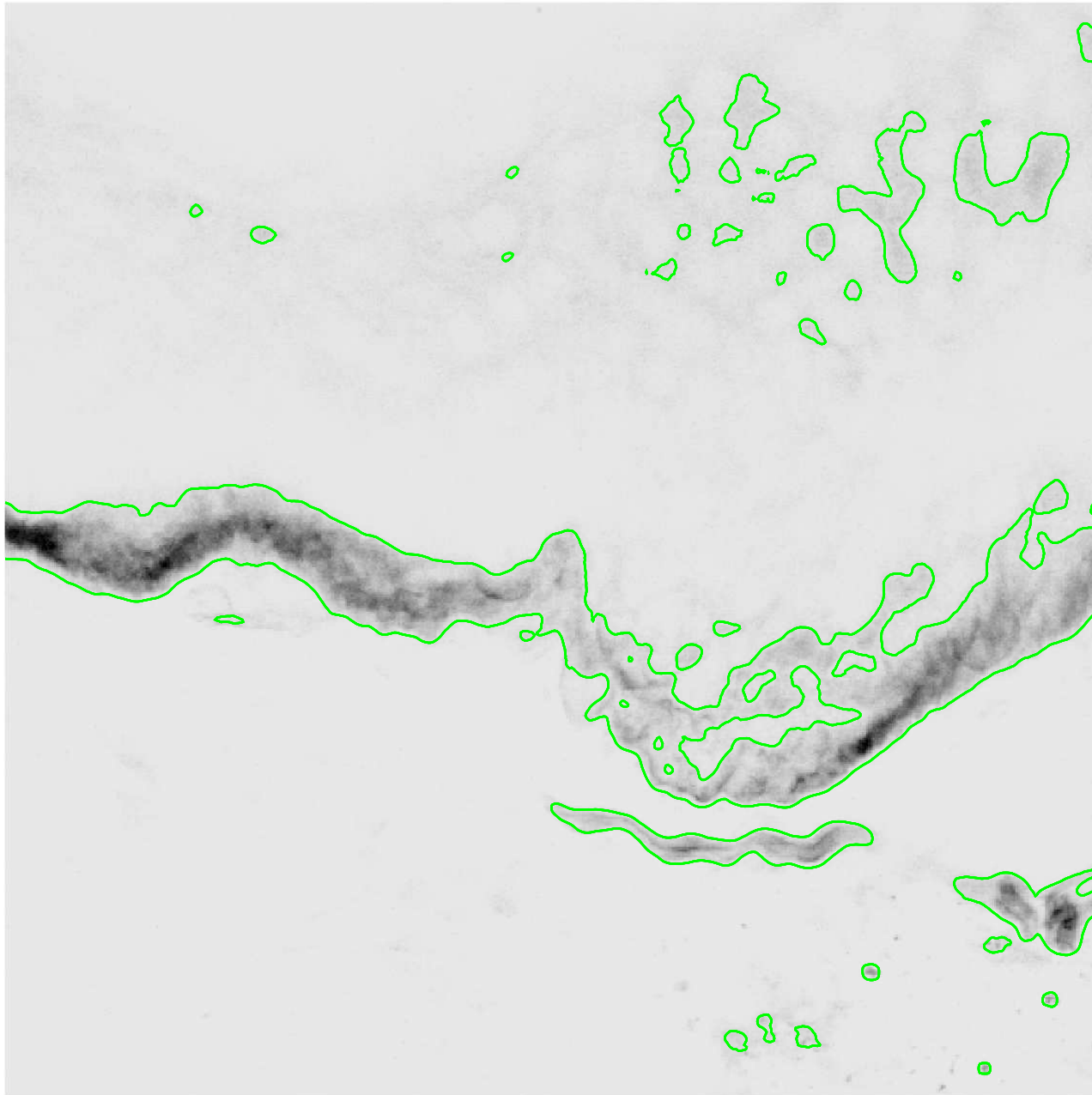
The Source-Extractor finds C-fibers that cross the membrane (in red).
Black boxes: Exclusion zone where Source-Extractor avoids bogus objects.

(2b) Application to C-fiber Skin-biopsies in Normal and Obese Patients



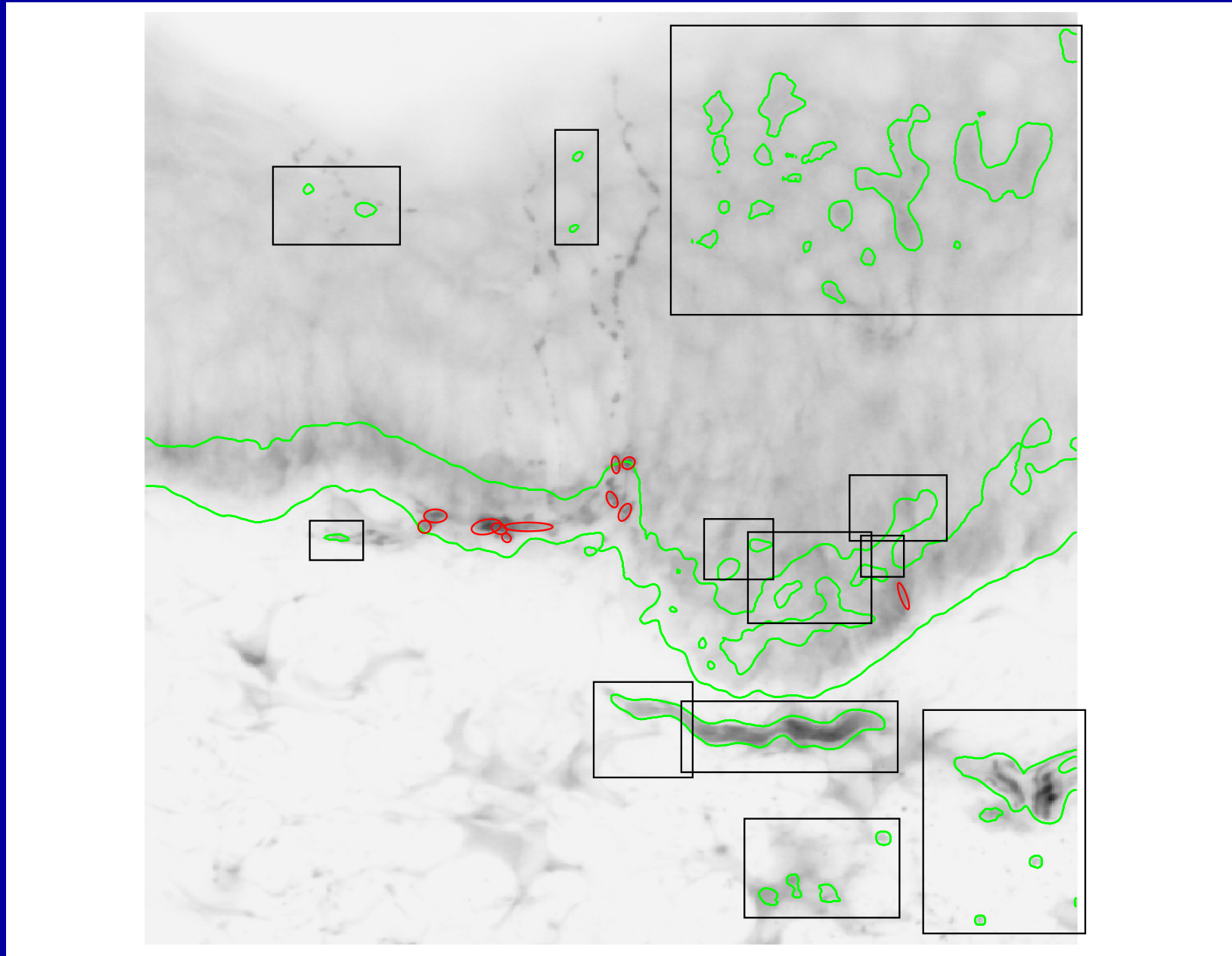
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(2b) Application to C-fiber Skin-biopsies in Normal and Obese Patients



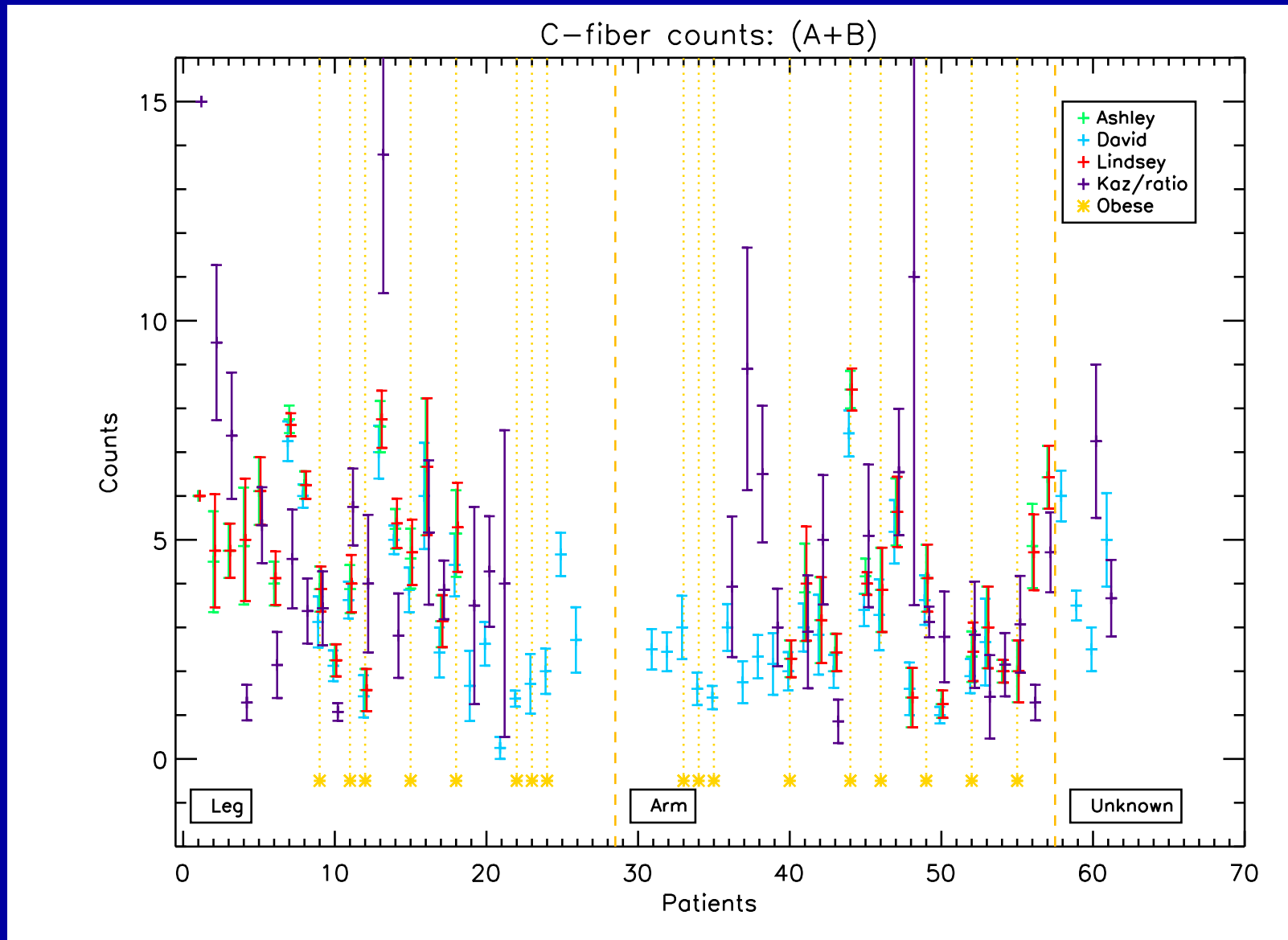
Different staining allows us to define the membrane and other parts of skin.

(2b) Application to C-fiber Skin-biopsies in Normal and Obese Patients



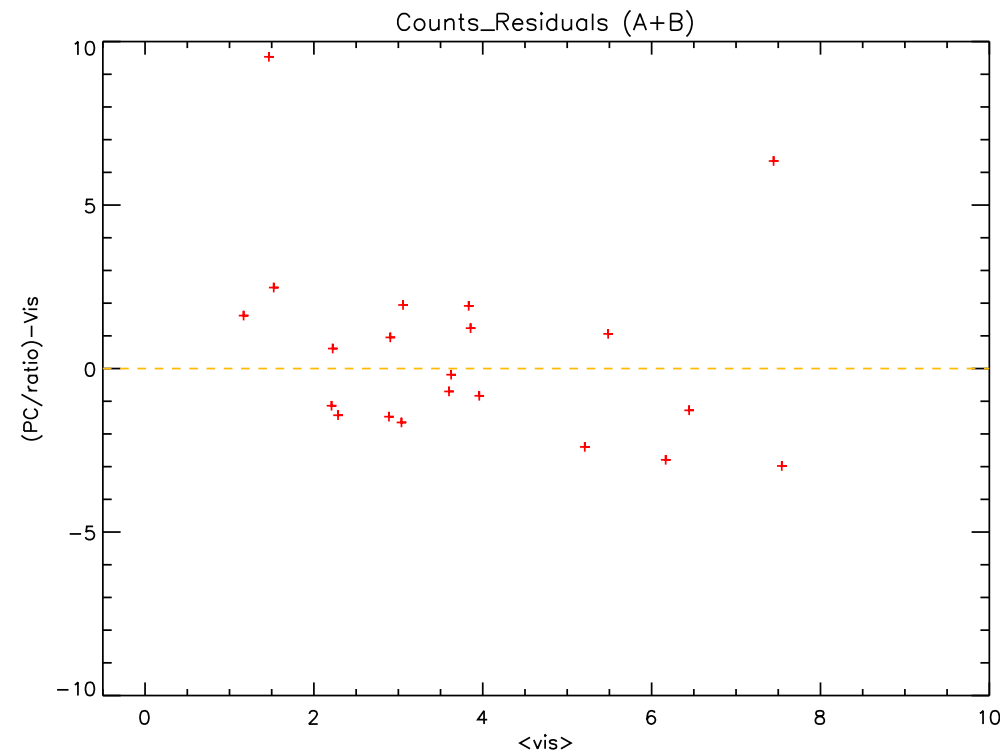
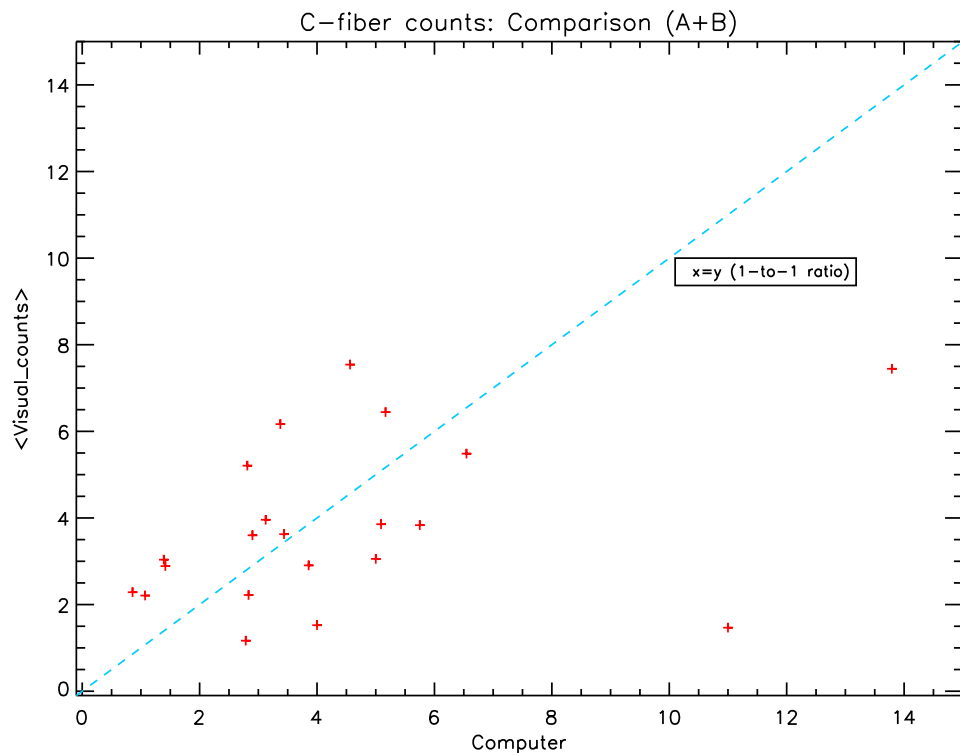
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(2b) Application to C-fiber Skin-biopsies



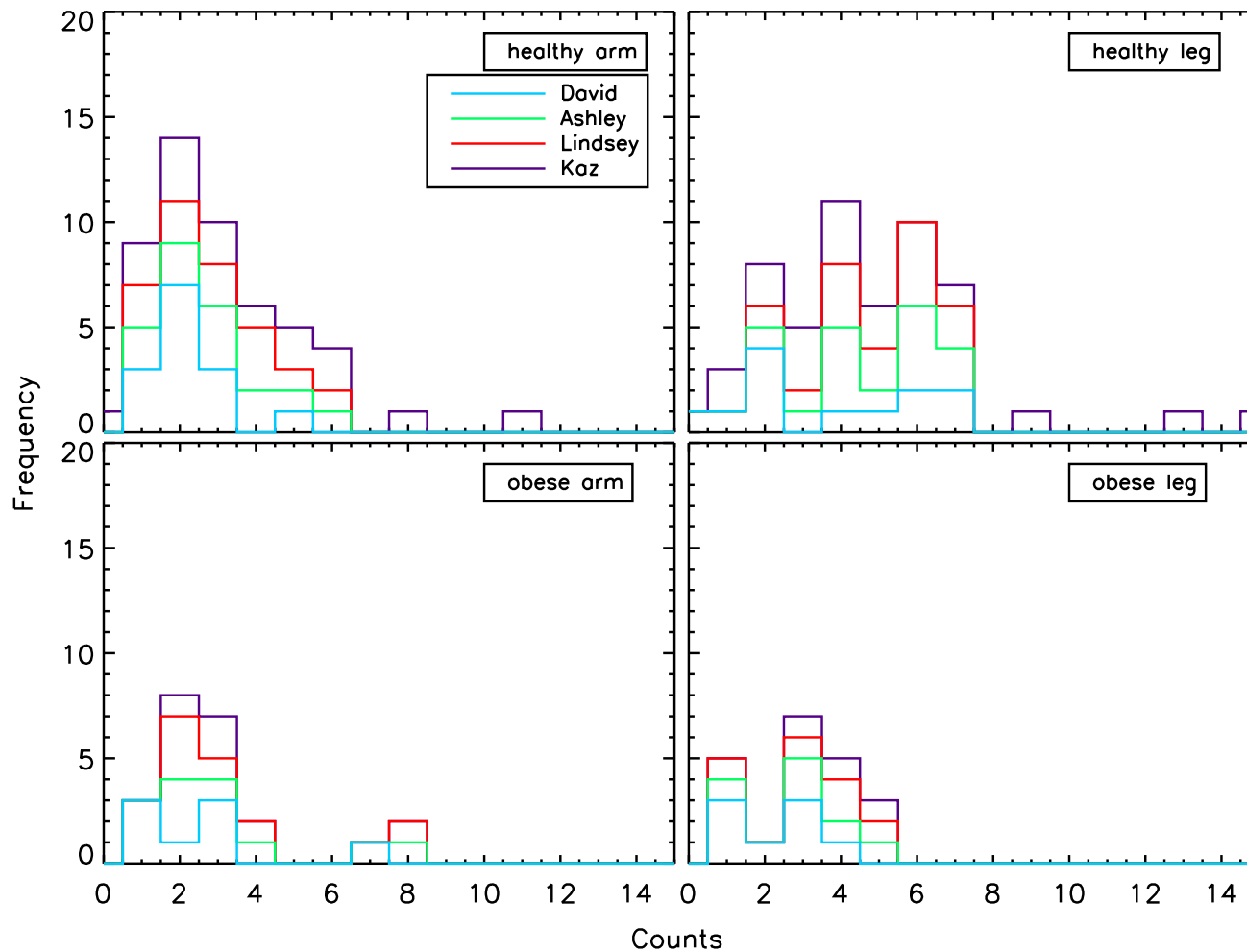
- Machine classification (black) follows average of 3 independent visual classifiers (different colors) quite well.
- Obese patients are marked in yellow, and have often lower C-fiber count.

(2b) Application to C-fiber Skin-biopsies



- Machine count correlates well with visual counts, with a few catastrophic failures where software breaks a C-fiber up in too many pieces.
- Average fragmentation is calibrated from expert visual classifiers.
- Catastrophic failures are corrected by hand — working on training the software to recognize and correct those as well.
- Dispersion between machine and visual classifiers is $\sigma \simeq \pm 1.5$ count.

(2b) Application to C-fiber Skin-biopsies



- Cumulative counts of both the machine and the visual classifiers show different distributions for normal and obese patients.
- Obese patients have in general lower C-fiber counts, although the differences between the two populations are not always huge.

REFERENCES

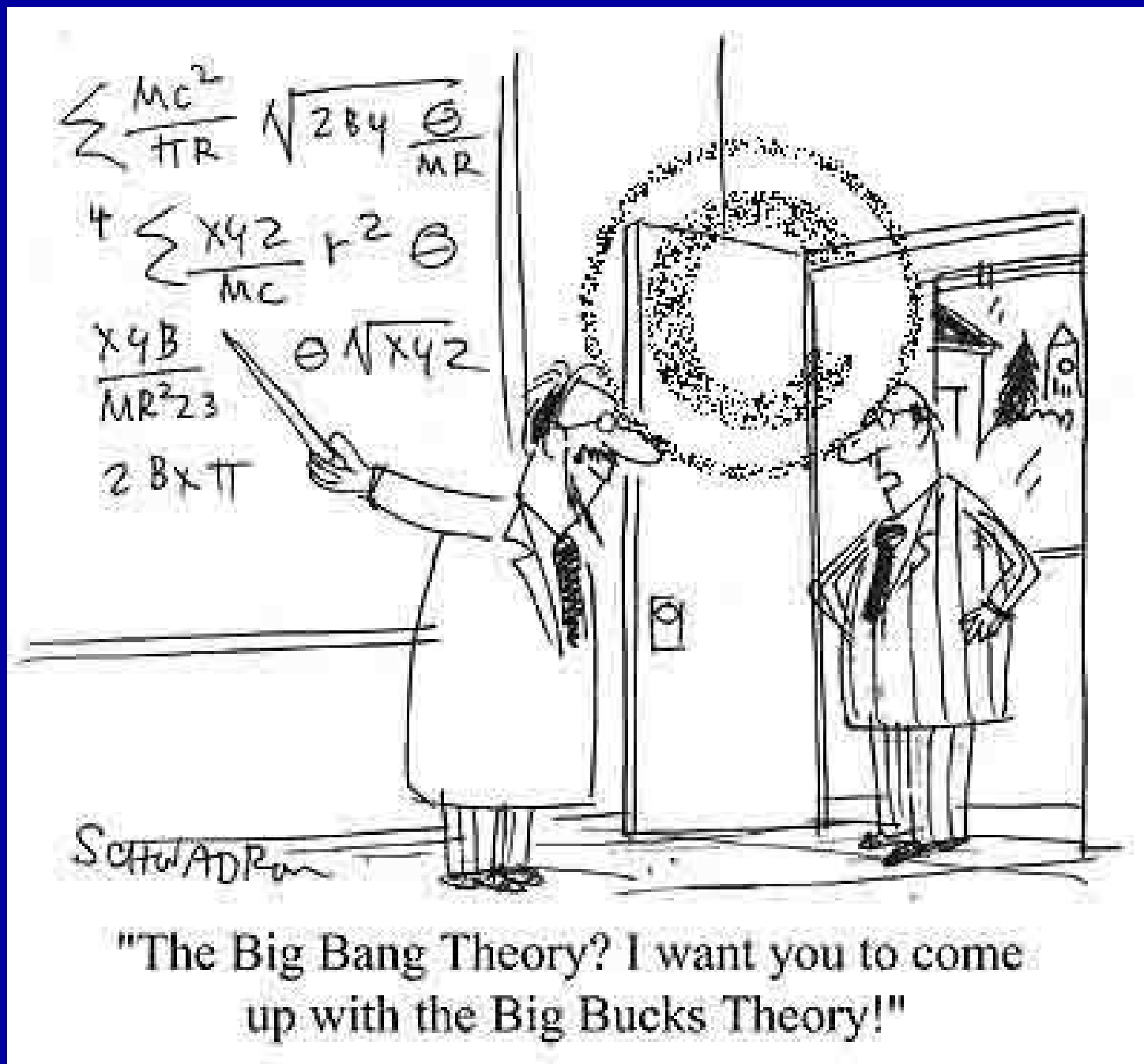
<http://www.asu.edu/clas/hst/www/medical/>

Casano, A. R., Brower, J. B., Baluch, D. P., Windhorst, R. A., Targovnik, J. H., & Herman, R. M. 2009, *International Journal of Obesity*, submitted (Jul. 2009) [<http://www.nature.com/ijo/index.html>] “Small Sensory Nerve Fiber Neuropathy in Morbidly Obese Subjects: Reduced Intra Epidermal Nerve Fiber Density”

Herman, R. M., Brower, J. B., Stoddard, D. G., Casano, A. R., Targovnik, J. H., Herman, J. H., & Tearse, P. 2006, *International Journal of Obesity*, 31, 226–235 [www.nature.com/ijo/journal/v31/n2/full/0803418a.html] “Prevalence of somatic small fiber neuropathy in obesity ”

Tamura, K., Mager, V. A., Burnett, L. A., Olson, J. H., Brower, J. B., Casano, A. R., Baluch, D. P., Targovnik, J. H., Windhorst, R. A., & Herman, R. M. 2009, *Journal of Neuroscience Methods*, resubmitted (Jul. 2009) [<http://ees.elsevier.com/jneumeth/>] “Semi-Automated Method of Analysis of Small Sensory Nerve Fibers in Human Skin-Biopsies”

SPARE CHARTS



Some cosmologists at ASU decided to step out of the box ...
(although it is hard to please the dean ...)

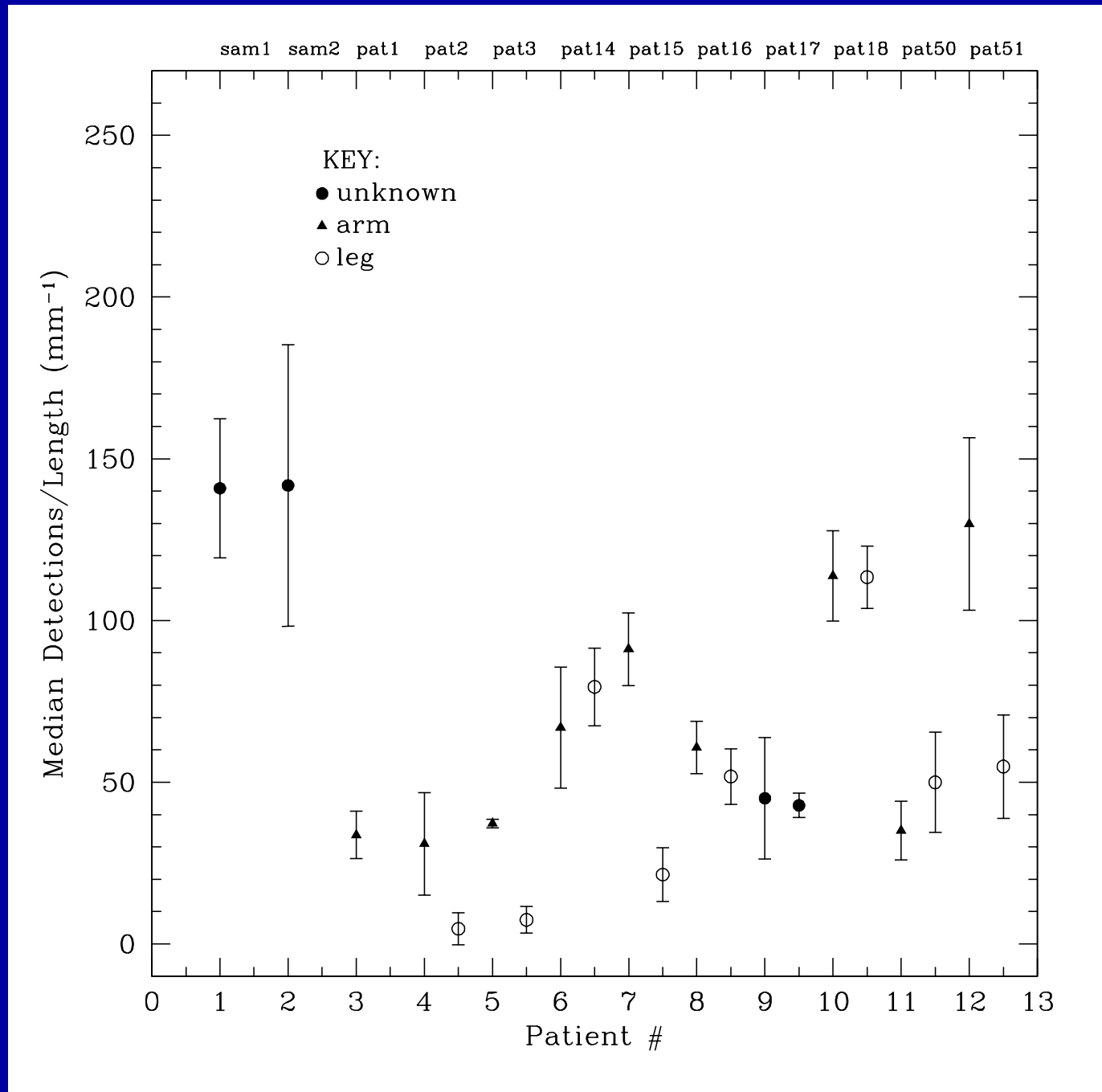
(1) The NASA Hubble Space Telescope connection

- [illegible]

⇒ Astronomers must be smart about using software to analyze what they see in Hubble images — we have absolutely no alternatives!.

This project of medical applications of Hubble galaxy classification software builds on a 40 year history of astronomical software development.

(2b) Application to C-fiber Skin-biopsies in Normal and Obese Patients



Number of C-fibers crossing membrane is directly counted for each patient.